Moments and Singular Perturbation

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September 19, 2025

1 Problem Setup

SYSTEM:

$$\dot{x} = A_{11}x + A_{12}z + B_1u \tag{1}$$

$$\varepsilon\dot{z} = A_{21}x + A_{22}z + B_2u \tag{2}$$

$$y = C_1x \tag{3}$$

No z term as otherwise we would have feedforward in the reduced model.
In state space form:

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B_1 \\ \frac{1}{\varepsilon}B_2 \end{bmatrix} u \tag{4}$$

SIGNAL:

$$\dot{w}_1 = S_{11}w_1 + S_{12}w_2 \tag{5}$$

$$\varepsilon\dot{w}_2 = S_{21}w_1 + S_{22}w_2 \tag{6}$$

REDUCED MODEL SIGNAL:

$$S_{22}w_2 + S_{21}w_1 = 0 \tag{7}$$

$$w_2 = -S_{22}^{-1}S_{21}w_1 \tag{8}$$

$$\dot{w}_1 = (S_{11} - S_{12}S_{22}^{-1}S_{21})w_1 \tag{9}$$

$$u = (L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \tag{10}$$

REDUCED MODEL OF THE SYSTEM:

$$A_{22}z + A_{21}x + B_2u = 0 \tag{11}$$

$$z = -A_{22}^{-1}(A_{21}x + B_2u) \tag{12}$$

$$\dot{x} = (A_{11} - A_{12}A_{22}^{-1}A_{21})x + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \tag{13}$$

FORWARD MOMENTS

$$M_g := \begin{bmatrix} C_1\Pi_{x1} & C_1\Pi_{x2} \end{bmatrix} \tag{14}$$

FORWARD SYLVESTER EQUATION FOR THE FULL SYSTEM:

$$\begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} + \begin{bmatrix} B_1L_{g1} & B_1L_{g2} \\ \frac{1}{\varepsilon}B_2L_{g1} & \frac{1}{\varepsilon}B_2L_{g2} \end{bmatrix} = \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ \frac{1}{\varepsilon}S_{21} & \frac{1}{\varepsilon}S_{22} \end{bmatrix} \tag{15}$$

FORWARD SYLVESTER EQUATION FOR THE REDUCED MODEL

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\tilde{\Pi} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) = \tilde{\Pi}(S_{11} - S_{12}S_{22}^{-1}S_{21}) \tag{16}$$

Can we show that as $\varepsilon \to 0$, the forward syvlester equation for the full system tends towards the forward sylvester equation for the reduced model?

2 Proof (I guess?)

Expanding out (14) we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1 L_{g1} = \Pi_{x1} S_{11} + \frac{1}{\varepsilon}\Pi_{x2} S_{21}, \tag{17a}$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1 L_{g2} = \Pi_{x1} S_{12} + \frac{1}{\varepsilon}\Pi_{x2} S_{22}, \tag{17b}$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2 L_{g1} = \Pi_{z1} S_{11} + \frac{1}{\varepsilon}\Pi_{z2} S_{21}, \tag{17c}$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2 L_{g2} = \Pi_{z1} S_{12} + \frac{1}{\varepsilon}\Pi_{z2} S_{22}. \tag{17d}$$

Multiplying (16c) and (16d) by ε we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1 L_{g1} = \Pi_{x1} S_{11} + \frac{1}{\varepsilon}\Pi_{x2} S_{21}, \tag{18a}$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1 L_{g2} = \Pi_{x1} S_{12} + \frac{1}{\varepsilon}\Pi_{x2} S_{22}, \tag{18b}$$

$$A_{21}\Pi_{x1} + A_{22}\Pi_{z1} + B_2 L_{g1} = \varepsilon\Pi_{z1} S_{11} + \Pi_{z2} S_{21}, \tag{18c}$$

$$A_{21}\Pi_{x2} + A_{22}\Pi_{z2} + B_2 L_{g2} = \varepsilon\Pi_{z1} S_{12} + \Pi_{z2} S_{22}. \tag{18d}$$

We now consider the ε series expansion of Π_{x1}, Π_{x2}, Π_{z1}, Π_{z2}. We can see that by multiplying (17a) and (17b) by ε, and setting ε to 0 we have that the zeroth order term for Π_{x2} is 0.
We then write:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1 L_{g1} = \Pi_{x1}^{(0)} S_{11} + \Pi_{x2}^{(1)} S_{21} + O(\varepsilon), \tag{19a}$$

$$A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2} = \Pi_{x1}^{(0)} S_{12} + \Pi_{x2}^{(1)} S_{22} + O(\varepsilon), \tag{19b}$$

$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2 L_{g1} = \Pi_{z2}^{(0)} S_{21} + O(\varepsilon), \tag{19c}$$

$$A_{22}\Pi_{z2}^{(0)} + B_2 L_{g2} = \Pi_{z2}^{(0)} S_{22} + O(\varepsilon) \tag{19d}$$

Each of the 4 equations in (18) are essentially polynomials in epsilon.
After equating coefficients, we can write $\Pi_{x2}^{(1)}$ as such (Assuming S_{22} is invertible):

$$\Pi_{x2}^{(1)} = \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2} - \Pi_{x1}^{(0)} S_{12}\right) S_{22}^{-1} \tag{20}$$

Substituting (19) into (18a) we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1 L_{g1} = \Pi_{x1}^{(0)} S_{11} + \left(A_{12}\Pi_{z2}^{(1)} + B_1 L_{g2} - \Pi_{x1}^{(0)} S_{12}\right) S_{22}^{-1} S_{21} + O(\varepsilon) \tag{21}$$

Rearranging we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1 L_{g1} = \Pi_{x1}^{(0)}\left(S_{11} - S_{12} S_{22}^{-1} S_{21}\right) + \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2}\right) S_{22}^{-1} S_{21} + O(\varepsilon) \tag{22}$$

Similarly, from (18c) we can substitute in an expression for $\Pi_{z1}^{(0)}$:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}A_{22}^{-1}\left(\Pi_{z2}^{(0)} S_{21} - A_{21}\Pi_{x1}^{(0)} - B_2 L_{g1}\right) + B_1 L_{g1} = \Pi_{x1}^{(0)}\left(S_{11} - S_{12} S_{22}^{-1} S_{21}\right) + \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2}\right) S_{22}^{-1} S_{21} + O(\varepsilon) \tag{23}$$

Rearranging we have:

$$\begin{aligned}
&\left(A_{11} - A_{12}A_{22}^{-1}A_{21}\right)\Pi_{x1}^{(0)} + \left(B_1 - A_{12}A_{22}^{-1}B_2\right) L_{g1} \\
&\quad + \left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2}\right) S_{22}^{-1}\right) S_{21} \\
&= \Pi_{x1}^{(0)}\left(S_{11} - S_{12} S_{22}^{-1} S_{21}\right) + O(\varepsilon)
\end{aligned} \tag{24}$$

Simplifying the middle (...) S_{21} term using (18d):

$$\left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - A_{12}\Pi_{z2}^{(0)} S_{22}^{-1} - B_1 L_{g2} S_{22}^{-1}\right) S_{21} \tag{25}$$

$$= \left(A_{12}A_{22}^{-1}\left(\Pi_{z2}^{(0)} S_{22} - A_{22}\Pi_{z2}^{(0)}\right) S_{22} - B_1 L_{g2} S_{22}^{-1}\right) S_{21} \tag{26}$$

$$= \left(A_{12}A_{22}^{-1} B_2 L_{g2} S_{22}^{-1} - B_1 L_{g2} S_{22}^{-1}\right) S_{21} \tag{27}$$

$$= -\left(B_1 - A_{12}A_{22}^{-1}B_2\right) L_{g2} S_{22}^{-1} S_{21} \tag{28}$$

Substituting (27) back into (23) and factorising we have:

$$\left(A_{11} - A_{12}A_{22}^{-1}A_{21}\right)\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) \tag{29}$$
$$= \Pi_{x1}^{(0)}\left(S_{11} - S_{12}S_{22}^{-1}S_{21}\right) + O(\varepsilon) \tag{30}$$

which is of $O(\varepsilon)$ error from (15). Thus as $\varepsilon \to 0$, (29) approaches (15), and the matrix $\Pi_{x1} \to \Pi_{x1}^{(0)} = \tilde{\Pi}$.

3 Next Steps

- Expression for Π_{x2} (Is it needed?)
- Higher order epsilon expansion?
- Assumptions, and rigour?

4 S_{11} and S_{22} are 0

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \tag{31a}$$
$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12}, \tag{31b}$$
$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2L_{g1} = \frac{1}{\varepsilon}\Pi_{z2}S_{21}, \tag{31c}$$
$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2L_{g2} = \Pi_{z1}S_{12}. \tag{31d}$$

Similar approach to before, we'll have $\Pi_{x2}^{(0)} = 0$.
Considering zeroth order terms:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x2}^{(1)}S_{21}, \tag{32a}$$
$$A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} = \Pi_{x1}^{(0)}S_{12}, \tag{32b}$$
$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2L_{g1} = \Pi_{z2}^{(0)}S_{21}, \tag{32c}$$
$$A_{22}\Pi_{z2}^{(0)} + B_2L_{g2} = 0 \tag{32d}$$

Assuming A_{22} is invertible:

$$\Pi_{z2}^{(0)} = -A_{22}^{-1}B_2L_{g2} \tag{33}$$

Assuming S_{12} is invertible, (33) $\to$ (32b):

$$\Pi_{x1}^{(0)} = (-A_{12}A_{22}^{-1}B_2L_{g2} + B_1L_{g1})S_{12}^{-1} \tag{34}$$

And $\Pi_{z1}^{(0)}$ can be extracted from (32c). Simulation results doesnt seem to agree. Might be implementation issue.

5 Thoughts

- Canonical form - is there a coordinate change that we can exploit?
- Making a family of approximations? what to do when A_{22} is not Hurwitz?
- Realisation?
- Controllability and observability conditions
- Is there a limit for the difference? $\Pi w_g - x$

6 Descriptor Systems

One thing to note that you could represent singular perturbation systems as a descriptor system. Scarlotti wrote a paper on singular systems and moment matching, but something different is that for us, our signal is also a singular perturbation system. Another thing to note, when you consider the scenario where S_{11} and S_{22} are 0, the signal is a pure integrator, but also when you set $\varepsilon = 0$, you have $S_21 w_1 = 0$ which is ambiguous. A reduced model for the sginal is not well defined. So a better approach might be to consider finite ε and then take the limit as $\varepsilon \to 0$.

6.1 New Definition of Moments

I think a better definition for the matrix Π is to consider it as a linear transformation of the signal w, such that the difference between Πw and x is governed by the open loop system dynamics. Then, the Sylvester equation becomes a condition that Π must satisfy for it to be a valid linear transformation. This allows us to move away from the frequency domain defintion of moments which relies on the system having a steady state response. Even for unstable systems, we can still compute a solution for the Sylvester equation when there is no steady state response.

$$\frac{d}{dt}(x - \Pi w) = A(x - \Pi w) \tag{35}$$

This way a nonlinear extension should be more straightforward, as Π will probably be some nonlinear operation on w.

7 Descriptor System Representation

Using a decriptor system representation:

$$\begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u \tag{36}$$

The signal is similarly:

$$\begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix} \begin{bmatrix} \dot{w}_1 \\ \dot{w}_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \tag{37}$$

Π is then defined as:

$$\begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix} \frac{d}{dt} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right) = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right) \tag{38}$$

The resulting Sylvester equation for the whole system then becomes:

$$A\Pi + BL_g = E\Pi E^{-1} S \tag{39}$$

where

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad E = \begin{bmatrix} I & 0 \\ 0 & \varepsilon I \end{bmatrix}, \quad S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad \Pi = \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix}, \quad L_g = \begin{bmatrix} L_{g1} & L_{g2} \end{bmatrix} \tag{40}$$

Setting S_{11} and S_{22} to 0, we have:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} \begin{bmatrix} L_{g1} & L_{g2} \end{bmatrix} = \begin{bmatrix} \Pi_{x1} & \frac{1}{\varepsilon}\Pi_{x2} \\ \varepsilon\Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} 0 & S_{12} \\ S_{21} & 0 \end{bmatrix} \tag{41}$$

The resuting 4 equations are then:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1 L_{g1} = \frac{1}{\varepsilon}\Pi_{x2} S_{21}, \tag{42a}$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1 L_{g2} = \Pi_{x1} S_{12}, \tag{42b}$$

$$A_{21}\Pi_{x1} + A_{22}\Pi_{z1} + B_2 L_{g1} = \varepsilon\Pi_{z1} S_{12} + \Pi_{z2} S_{21}, \tag{42c}$$

$$A_{21}\Pi_{x2} + A_{22}\Pi_{z2} + B_2 L_{g2} = \Pi_{z1} S_{12} + \Pi_{z2} S_{22}. \tag{42d}$$

Using expansion in ε, we can find the where the partitions of Π will converge to as $\varepsilon \to 0$. When S_{22} is 0, it is singular and the reduced model for the signla is undefined. The significance of this representation is that, even when there is no reduced model fot the signal, we can still find a meaningful Π that relates the system and the signal.